

Evolutionary Computing (X_400111)

Standard Assignment, Task II: Neuroevolution

Send the Best, the Worst, or Anyone?

Emigrant Selection in Island-Model Neuroevolution
of a Walking Robot

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ABSTRACT

We evolve the 212 weights of a 16-8-4-8 neural network that makes the eight-hinge spider_8 robot of the ARIEL framework walk 2 m to a target on the OlympicArena within 15 s. Our research question is how the emigrant-selection policy of a four-island ring model (migrating the best, the worst or random individuals, or not migrating at all) affects the speed and the outcome of evolution. Over 20 paired seeds of 12,000 evaluations each, every evolutionary algorithm beats random search by far (final fitness 1.13–1.24 against 2.49; paired t -tests, Holm-corrected $p < 10^{-9}$). Between the policies the differences are small: migrating the best or random individuals probably converges faster than isolated islands (posterior probability 99% and 98% on the area under the convergence curve), migrating the worst barely does (76%), and no difference between the evolutionary algorithms is significant after correcting for the 15 pairwise comparisons, in speed, final fitness or on 20 unseen arenas. By end results alone, a single population generalises best and random migration ends fittest and most consistently, but neither lead is significant. The 15 s limit binds: given 20 s, 39 of 120 best brains reach their target instead of 19.

KEYWORDS

Neuroevolution, island model, migration policy, evolutionary robotics, legged locomotion

1 INTRODUCTION

Evolutionary robotics designs robot controllers by evolving them in simulation [17]. A common choice is to fix the topology of a neural network and evolve its weights [10]: a continuous search space of hundreds of dimensions, rugged and multimodal, where a population easily converges early on a mediocre gait.

Island models are a standard defence against such premature convergence. The population is split into sub-populations (islands, or demes) that evolve separately and exchange a few individuals from time to time (migration) [1, 3, 22]. Isolation lets the islands explore different regions; migration spreads good genetic material. How an island model behaves depends on the topology, the interval and size of migration, how emigrants are selected and whom they replace [20]. Cantú-Paz [4] showed that choosing emigrants by fitness raises the selection pressure of the whole algorithm, which speeds up convergence but risks taking every island to the same solution. Those results concern genetic algorithms on benchmark functions. Neuroevolution adds a twist: two networks that solve a task can order their hidden neurons differently (the competing-conventions problem [18]), so crossing a native with an immigrant from another lineage can produce a broken child.

Whether migrating the best is still the right choice for evolving a walking robot is therefore an open, practical question.

We study it on the task of the assignment: evolve the weights of a neural-network controller so that a fixed robot body walks from its spawn point to a target. Our research question is:

RQ. How does the emigrant-selection policy of an island-model EA (migrating the *best*, the *worst* or *random* individuals, or *no* migration) affect the convergence speed and the final quality of evolved locomotion controllers?

Following Cantú-Paz [4], we expect: **H1**, migration of any kind converges faster than isolated islands; **H2**, migrating the best converges fastest, but ends at an equal or worse plateau than random migration, because the islands lose their differences; **H3**, migrating the worst behaves close to no migration. We compare the four policies with a standard single-population EA and random search on 20 paired seeds each, with a migration interval chosen in a preliminary study, and follow up with longer test walks.

2 METHODS

2.1 Task, body and world

The body is spider_8 from ARIEL’s John Set: a core with four legs of two hinges each, eight position-controlled motors in total. The world is the OlympicArena: a flat start, then from $x = 0.5$ m a strip of centimetre-high bumps on which the target lies at (2, 0) m (Figure 1). The robot spawns at the origin with ARIEL’s default spawn, and each walk lasts 15 s, as in the template; it ends early when the core comes within 0.1 m of the target. The network is queried at 50 Hz (every 10 physics steps of 2 ms); the motors’ servos move the joints in between.

Both body and world are free choices that stay fixed for the whole study; we chose them in pilots. Our first body, spider_16, cannot lift itself: holding its 4 kg up needs about 5 N m per leg, while every John Set motor is limited to 0.66 N m. On the rugged world the mean slope along the path is 25° and no body we tried got far; on the flat world spider_8 solves the task within about 2,000 evaluations, leaving no room for conditions to differ. The OlympicArena is learnable but not trivial.

ARIEL draws the bumpy strip with unseeded noise, so every build is a new arena. Each seed therefore builds one arena once and every condition with that seed walks the same saved arena. This makes the fitness deterministic and the comparison between conditions paired by seed. Generalisation is tested afterwards on 20 fresh arenas (Section 3).

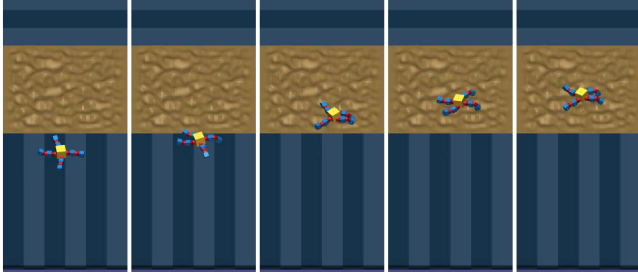


Figure 1: An evolved brain (standard EA, seed 11) walking up the OlympicArena, seen from above, at 0.2, 2.5, 5, 7.5 and 10.2 s. It starts on the flat part (bottom) and reaches the target on the bumpy strip after 10.3 s.

Table 1: The controller’s 16 inputs, all scaled to about $[-1, 1]$.

Input	Size	What it tells the controller
Joint angles	8	where each hinge is now
Clock	2	sin and cos of a 1 Hz beat, for rhythm
Target	3	distance / 2 m; sin, cos of its bearing
Tilt	3	the up direction seen from the core

2.2 Controller

The controller is a fully connected feed-forward network with two hidden layers of 8 and 4 tanh neurons and 8 tanh outputs, each scaled to a target angle in $[-\pi/2, \pi/2]$ for one hinge. With a bias on every neuron it has 212 weights, which form the genome. Its 16 inputs are listed in Table 1. The network has no memory, so it gets a clock to walk to; the clock is a fixed input signal, not an oscillator inside the controller, so the controller is not a CPG. The target is given relative to the robot’s heading rather than as an absolute position, so that “the goal is 1 m away, 30° to the left” means the same everywhere on the arena.

Why no vision. Our first controller also had distance rays to the terrain, designed for the rugged world’s half-metre hills. On the OlympicArena a ray reads up to 3 m while the bumps are about ± 2 cm, so a bump moved a ray input by less than 0.01, and five, three, one or no rays did equally well (Table 2). We removed vision, which also shrank the genome.

Why 16-8-4-8. Of six shapes (Table 2), the bottleneck of four was the only one better than a single layer of 16 on every tuning seed and on unseen arenas: driving eight motors through four shared signals pushes the legs towards coordinated patterns, with half the weights.

2.3 Fitness function

The assignment’s fitness is the planar distance d_T between the core’s final position and the target. We started with exactly that, and nothing walked: the best networks of our first runs wiggled in place, their joint commands within ± 0.13 of their range. Penalising the core for touching the ground and for being upside down made the robots move, but on video they crouched (the core 1.6 cm above lying height, although the body can hold it 6 cm up) and kept one

leg tucked away ($5-9^\circ$ of movement against $17-77^\circ$ for the other legs). The distance alone gives evolution no reason to walk well before it walks at all, so we added terms for the mechanics of walking, and minimise

$$f = d_T + 0.5 \bar{d} + 1.0 c + 1.0 u + 1.0 \ell + 0.5 w, \quad (1)$$

where every term is measured at each 50 Hz update and lies in $[0, 1]$ except the distances:

- **Stand** (c, ℓ): c is the share of the walk in which the core touches the ground, and ℓ is the mean shortfall of the core’s lift above its lying height from a 4 cm line, $\text{clip}(1 - \text{lift}/4 \text{ cm}, 0, 1)$. The body can hold itself 6 cm up, so 4 cm is demanding but reachable, and every millimetre up to the line pays.
- **Use all legs alike** (w): just as people use both legs about equally, each leg should do about a quarter of the work. With W_i the positive motor work of leg i , $w = 1 - \min_i W_i / \bar{W}$: 0 when every leg works alike, 1 when one leg is idle. We first measured how much each leg moves, but a leg can move without pushing, and a leg that only props the body up carries its share of the weight; only the motor work caught such a lazy leg (22 J against up to 40 J for the others).
- **Stay upright** (u): the share of the walk spent upside down.
- **Be fast** ($\bar{d}$): the distance to the target averaged over the whole 15 s, counting 0 after arrival, so that closing the distance early, and arriving early, lower the fitness.

The third mechanic of walking, **rhythm**, is not a fitness term: it is provided by the clock input. A robot that holds still scores 5.0 ($d_T = \bar{d} = 2$, core on the ground, nothing lifted), and one that arrives at once while standing tall and using all legs alike scores 0. A typical evolved brain (no migration, seed 10) scores $0.29 + 0.5 \cdot 1.12 + 0 + 0 + 0.09 + 0.5 \cdot 0.11 = 1.00$: the two distance terms dominate, and the gait terms decide between walkers that get about equally far. This is behavioural shaping: it prescribes part of *how* to walk, which the assignment allows if the metric is clearly defined. All conditions use the same fitness, and we report the plain distance as well.

2.4 The evolutionary algorithm

The genome is a fixed-length vector of real numbers, so we use a real-coded genetic algorithm: Gaussian mutation explores around good weights, and recombination can combine good neurons found in different individuals. The island model is both the object of our study and a known guard against premature convergence on multimodal landscapes.

Representation and variation. A genome is the vector of 212 real weights, initialised from $\mathcal{N}(0, 0.5^2)$. Parents are chosen by tournaments of three within an island [9]. With probability 0.9 a child is made by *neuron-level uniform crossover*: for every hidden neuron, a fair coin decides from which parent it inherits all of that neuron’s incoming weights and bias (the second hidden layer’s neurons also take their outgoing weights); each output bias comes from a random parent. Otherwise the child copies one parent. Every child is then mutated by adding $\mathcal{N}(0, 0.05^2)$ noise to every weight, with ARIEL’s Gaussian mutation operator. Keeping neurons whole is the classic response to the competing-conventions

problem [15, 16]: the child inherits working units rather than a mixture of two units that may play different roles.

Survivor selection. Generational with elitism: each island keeps its two best individuals unchanged and replaces the other 18 by children. As the arena is fixed, the fitness is deterministic and elites are not re-evaluated.

Island model. Four islands of 20 individuals form a ring. Every 20 generations each island copies two emigrants to the next island, where they replace that island’s two worst individuals. The emigrant-selection policy is the variable of the RQ: *best* (the two fittest), *worst* (the two least fit), *random* (two chosen uniformly) or *none* (no migration, four isolated islands). Everything else is identical across conditions.

Baselines. The *standard EA* is the same algorithm with one population of 80 and 8 elites, so only the population structure differs. *Random search* draws every new genome from the initial distribution and keeps the best, at the same budget, as the assignment requires.

Budget. Each run stops after 12,000 evaluations (80 initial and 72 children per generation: about 166 generations). Instead of a plateau rule, which never fired before the budget ran out in our pilots, we chose the budget from where the pilot curves flattened: 95% of the progress of a long run came within 5,000 evaluations and its last improvement at 8,800.

2.5 How the settings were chosen

Table 2 lists every setting with its justification. Settings were changed by controlled comparisons with a decision rule fixed beforehand: a new value replaces the current one only if its best fitness after 6,000 evaluations is better on all three tuning seeds and on the mean. The tuning seeds (100–102) and their arenas are disjoint from those of the final experiment (10–29), and tuning used the random emigrant policy, which favours none of the policies we compare. Only one tuned value changed: crossover probability 0.9 (mean best fitness 1.369 against 1.800 at 0.5). The number of migrants, the topology and the replacement were not tuned: they set the context of the RQ and take common values from the literature [4]. The interval started at the common 10 generations; a preliminary study (Section 4.4) set it to 20, a design choice rather than a tuning win: at 20 the islands stay distinct enough for the emigrant policy to matter.

3 EXPERIMENTAL SETUP

Conditions. Six conditions: the island EA with the *best*, *worst*, *random* and *none* policies, the standard EA and random search. Each runs on seeds 10–29, 120 runs in all; a seed fixes the EA’s random numbers and the arena, and all conditions with the same seed share that arena, which pairs the runs.

Preliminary study. A first version of this experiment ran the same six conditions on five seeds (0–4), migrating every 10 generations, and then the *best* policy every 5, 20 and 50 generations on the same seeds and arenas (Section 4.4). It set the interval of the final experiment to 20. Because the interval was chosen on seeds 0–4, the final experiment uses 20 fresh seeds and arenas.

Metrics. Lower is better for all. (1) The best-so-far fitness against evaluations, and its area under the curve (AUC), which rewards being good early. (2) Convergence speed: the evaluations until the best fitness first drops below 1.6. This threshold separated fast from slow runs in tuning, where no untuned run reached it. (3) The final best fitness after 12,000 evaluations, and the closest any brain got to the target. (4) Generalisation: each run’s best brain walks 20 fresh arenas it never saw, and we report the mean distance left.

Statistics. Every comparison is paired by seed. A Friedman test across the six conditions asks whether they differ at all. Then every pair of conditions (15 pairs) is compared on the final fitness, the AUC and the unseen distance with a two-sided paired test, Holm-corrected across the pairs of each metric [12]. The test was fixed before any result existed: if the per-seed differences of every pair look normal (Shapiro–Wilk, Holm-corrected, $\alpha = 0.05$ [19]), the metric gets the paired *t*-test; otherwise the Wilcoxon signed-rank test [23], the usual choice for comparing evolutionary algorithms [6, 7], which needs no normality and keeps 95% of the *t*-test’s efficiency when the data are normal [11]. We test normality rather than assume it because the arenas differ in difficulty: in the preliminary study the seed explained 37% of the EAs’ variance in final fitness and the condition 14%, so a hard arena can make one seed’s difference an outlier. For every pair we also give the probability that one condition has the lower true mean, from a Bayesian paired *t*-test with a flat prior [2], with differences within ± 0.05 counting as practically equal.

End results. After seeing the results we added an exploratory analysis of where the runs end, with reaching the target first (Section 4.3). Every best brain also walks its own arena for 15, 20, 30 and 60 s and the unseen arenas for 30 s. Yes/no arrivals are compared with Cochran’s *Q* [5] and exact McNemar tests [14], the rest with the rule above, Holm-corrected over the ten pairs of EAs.

3.1 Compute

Table 3 counts every run of the study. An evaluation is mostly one 15 s walk of 7,500 physics steps of 2 ms; with the unseen tests (10,809 walks) the study simulated about 14,900 hours of walking. Each run spread its walks over 10–20 worker processes, so the 57 machine hours are at most roughly 600 CPU-core hours. The final experiment ran on two machines at once, split by seed (an Apple M3 Pro and an Intel i7-12700H, about 7–8 minutes per run), and took 8.4 hours. Smoke tests, scratch diagnostics, overwritten runs and one 0.36 M-evaluation showcase run outside the study are not counted. The code builds on ARIEL without changing it, and one script per experiment runs all its seeds and conditions. [TODO: name the scripts or the repository in one sentence]

4 RESULTS

4.1 Convergence and final fitness

Figure 2, Table 4 and Figure 3 summarise the final experiment. The conditions differ (Friedman $\chi^2 = 46.6$ on the final fitness and 47.5 on the AUC, both $p < 10^{-8}$), and the difference is random search: every EA beats it on every metric (Holm-corrected $p \leq 0.007$; posterior probability 100%). Random search never reached a fitness of 1.6 and came no closer than 1.21 m to the target. Following the rule

Table 2: Every setting of the final EA and how it was chosen. “Tested” rows were decided by the rule of Section 2.5 (3 seeds, 6,000 evaluations); best fitness shown as the mean over the seeds, lower is better.

Component	Setting	How it was chosen
Network	16-8-4-8, tanh, 212 weights	tested against 16-8-8, 16-16-8, 16-32-8, 16-8-8-8, 16-16-16-8: only shape better on every seed (1.203 vs 1.339)
Inputs	16, no vision	10 → 5 rays (pilot); 5 vs 3, 1, 0 rays tested: no difference, rays carry <0.01 of signal
Parent selection	tournament, $k = 3$	$k = 2$ and 5 tested: inconsistent (1.877, 1.612 vs 1.800)
Crossover	neuron-level uniform	tested against weight-level uniform (1.416), BLX-0.5 (1.168, better on 2 of 3 seeds) and headless chicken (2.207) vs ours (1.203)
Crossover probability	0.9	0, 0.5 and 0.9 tested: 0.9 better on every seed (1.369 vs 1.800; 0 gave 2.063)
Mutation	Gaussian, $\sigma = 0.05$, every weight	pilot over $\sigma \in \{0.05, 0.1, 0.2\}$; then 0.02, 0.1, and 10% of weights with $\sigma = 0.16$ tested: none better
Mutation-step control	none	a rule doubling σ on stagnating islands tested: 1.369 vs 1.348 without, dropped
Survivor selection	generational, 2 elites per island	1 and 5 elites tested: inconsistent (1.791, 1.811 vs 1.800)
Population	4 islands $\times$ 20	4×10 and 4×40 tested: inconsistent (2.020, 1.693)
Migration	ring, every 20 generations, 2 copies, replace worst	literature [4], interval from the preliminary study (Section 4.4): every 20 kept the islands apart
Budget	12,000 evaluations (≈ 166 generations)	pilot curves: 95% of the progress by 5,000, last gain at 8,800
Walk	15 s, ends at the target	the template’s length; 20 s training walks tested: no better on unseen arenas

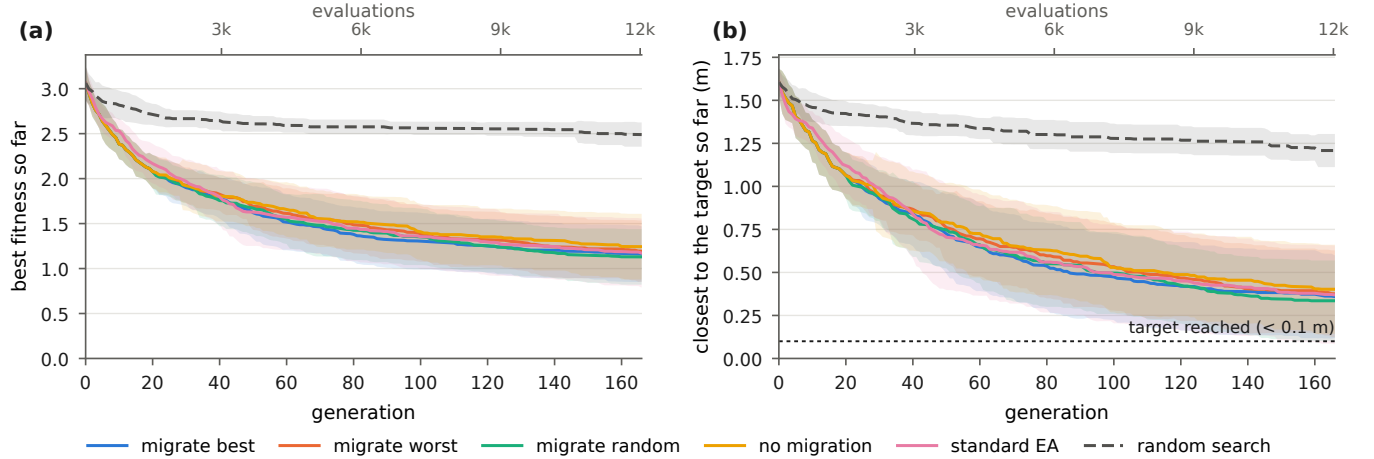


Figure 2: Best-so-far fitness (a) and closest distance to the target (b) against generations (top axis: evaluations), for every condition: the mean over 20 seeds, with a band of ± 1 standard deviation. Lower is better.

Table 3: Compute of the whole study, from the run logs: runs, evaluations (mostly one 15 s simulated walk each) and machine hours (run times added up).

Phase	Runs	Evaluations	Machine h
Development (task, body, fitness)	69	0.80 M	20.7
Tuning (settings, 3 seeds)	111	0.83 M	14.5
Preliminary study (5 seeds)	45	0.54 M	6.4
Final experiment (20 seeds)	120	1.44 M	15.4
Total	345	3.62 M	57.0

fixed in advance, the final fitness and the AUC were tested with paired t -tests, as no pair’s differences rejected normality, and the unseen distance with Wilcoxon tests: one isolated-islands brain fell

off 3 of its 20 unseen arenas within 15 s, which count as 10 m from the target, and skewed the differences of most pairs with *none*.

Between the EAs the differences are small. By the AUC, migrating the best or random individuals is probably faster than isolated islands (posterior 99% and 98%; best minus none -0.086 , 95% CI -0.152 to -0.020 , $d_z = -0.61$, $p = 0.014$), but neither survives the correction for 15 comparisons (Holm $p = 0.14$ and 0.27). Migrating the worst is the slowest migration policy: best and random are faster with probability 90% and 94%, and it is only 76% faster than none. Migrating the best still reached fitness 1.6 first (4,076 against 4,501 without migration), with a far smaller lead than in the preliminary study. In the final fitness, random migration ended best (1.130; 96% better than none, 73% than best), and the standard EA sat among the islands (1.171; 36–71%). No paired test between two EAs is significant on any metric (final fitness: all Holm $p \geq 0.81$).

Table 4: Results of the final experiment after 12,000 evaluations, mean $\pm$ standard deviation over 20 seeds (lower is better). “Closest” is the closest any brain of the run got to the target; “to 1.6” counts only the runs that got there. P(faster than no migration) is the probability, from a Bayesian paired t -test over the 20 seeds, that the condition’s mean AUC is lower (it improves faster) than that of the isolated islands; 50% would mean no evidence either way.

Condition	Final fitness	Closest (m)	Evaluations to 1.6	AUC	Unseen arenas (m)	P(faster than no migration)
Migrate best	1.162 ± 0.307	0.359 ± 0.239	4,076 $\pm$ 1,867 (18/20)	1.562 $\pm$ 0.243	0.932 ± 0.201	99%
Migrate worst	1.194 ± 0.341	0.376 ± 0.253	$4,891 \pm 2,404$ (17/20)	1.620 ± 0.234	0.915 ± 0.182	76%
Migrate random	1.130 $\pm$ 0.309	0.335 $\pm$ 0.234	$4,576 \pm 1,980$ (18/20)	1.570 ± 0.244	0.850 ± 0.194	98%
No migration	1.243 ± 0.365	0.402 ± 0.259	$4,501 \pm 1,489$ (15/20)	1.648 ± 0.252	1.005 ± 0.338	–
Standard EA	1.171 ± 0.383	0.369 ± 0.287	$4,713 \pm 2,860$ (17/20)	1.610 ± 0.289	0.825 $\pm$ 0.221	66%
Random search	2.490 ± 0.135	1.208 ± 0.096	never (0/20)	2.612 ± 0.068	1.406 ± 0.118	0%

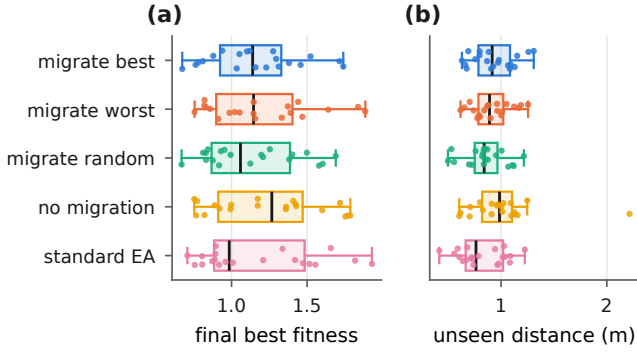


Figure 3: Every seed’s final best fitness (a) and distance left on the 20 unseen arenas (b), per condition: boxes show the quartiles and the median, dots are the 20 seeds. Random search (about 2.5 and 1.4 m) is left out for scale.

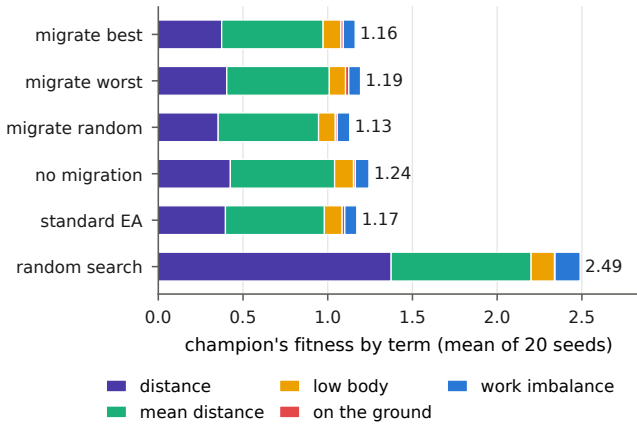


Figure 4: The final champions’ fitness split into the weighted terms of Equation 1, mean over 20 seeds; “mean distance” is $0.5 \bar{d}$. No champion was ever upside down.

Table 5: End results of the five EAs (exploratory). Own target: best brains that reach the target of their training arena in 15 s and within 30 s, of 20. Unseen: share of the 400 walks on unseen arenas that reach the target within 30 s (none does within 15 s), mean fitness there over 15 s, and walks in which the robot fell off the arena within 30 s (a fall counts as 10 m from the target in the distances).

Condition	Own target		Unseen arenas		
	15 s	30 s	reached	fitness	fell off
Migrate best	5	10	6.0%	1.886	17
Migrate worst	1	6	6.5%	1.854	7
Migrate random	4	7	10.2%	1.780	7
No migration	4	7	7.8%	2.010	6
Standard EA	5	9	12.0%	1.739	1

4.2 What the fitness rewards

In every EA’s champion the speed term $0.5 \bar{d}$ is about half of the fitness (0.58–0.62) and larger than the final distance (0.35–0.43; Figure 4). The gait terms add only 0.18–0.20: no champion is ever upside down and the core hardly touches the ground ($c \leq 0.02$ on average). The posture terms shaped the gait early and then left the ranking to the distances.

4.3 End results: which algorithm is best?

Speed matters only when compute is short, so we also asked which algorithm ends best, with reaching the target first (Table 5). No end result separates the EAs significantly: their arrival rates on the own arena do not differ (Cochran’s Q , $p = 0.41$ at 15 s and 0.63 at 30 s), nor do their unseen arrivals or unseen fitness (Wilcoxon, all Holm $p \geq 0.36$). Read descriptively, the standard EA generalises best: it reaches the most unseen targets (12% of its walks against 6–10%), has the best unseen fitness and fell off an arena once in 400 walks, against 6–17 times for the islands. Random migration ends with the best training fitness and is the steadiest: in final fitness it was never the worst of the five on any seed, while the standard EA is all-or-nothing, the best on 9 seeds and the worst on 7 (Figure 3a). Migrating the best brings the most brains to their own target but reaches the fewest unseen targets and falls off most often. All brains specialise to their arena: they end 0.35–0.43 m from their own target after 15 s, against 0.83–1.01 m on unseen arenas. And the 15 s limit

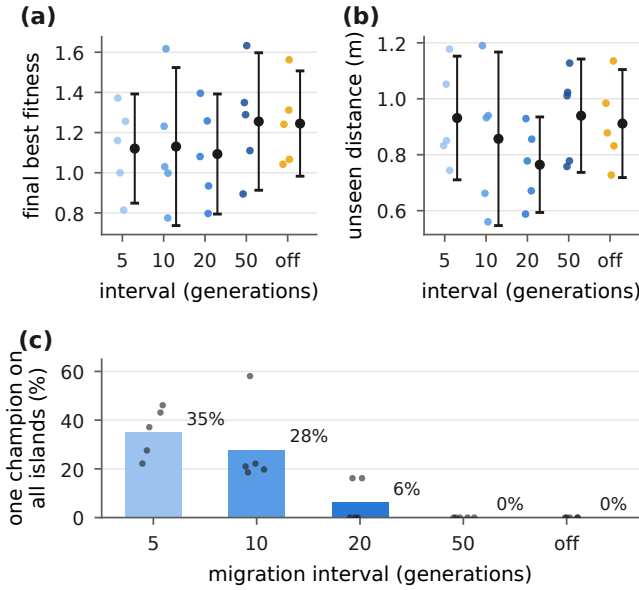


Figure 5: Preliminary study: the *best* policy migrating every 5, 10, 20 or 50 generations, or never (“off”), on seeds 0–4. (a) Final best fitness and (b) distance left on 20 unseen arenas: dots are seeds, black the mean with its 95% interval. (c) Share of generations in which all four islands hold the same champion; dots are seeds.

binds: 19 of the 120 best brains reach their own target within 15 s, 39 within 20 s; the others stall, most of them 0.1–1.0 m away, and more time hardly helps.

4.4 Preliminary study: how often to migrate

Figure 5 shows a sweet spot: every 5, 10 or 20 generations is equally fast (AUC probabilities 42–58% between them) and faster than every 50 (93–99%) or never (96–100%); three migrations in 166 generations are too few to matter. The more often the best migrate, the sooner one champion holds all islands: in 35% of generations at every 5, 28% at 10 and 6% at 20. Every 20 kept the islands apart and had the best final fitness (1.093) and unseen result (0.76 m), though only modestly (58% and 77% against every 10), so we chose it for the islands’ diversity. In the final experiment one champion held all four islands in 4.3% of the generations.

5 DISCUSSION

Answer to the RQ. With 20 seeds, the emigrant-selection policy has at most a small effect on how fast the islands improve and none we can detect on where they end. H1 is partly supported: migrating the best or random individuals is probably faster than isolated islands (99% and 98%; a medium effect for the best, $d_z = -0.61$), but not significantly after correcting for 15 comparisons, and migrating the worst barely helps (76%). H2 fits the data, weakly: migrating the best is the fastest by the AUC, though tied with random (59%), and ends at a slightly worse plateau than random migration (73%), as the selection-pressure argument predicts [4]. H3 is supported in direction: migrating the worst is the slowest migration policy and

the closest to no migration. By end results, no algorithm is reliably best either (Section 4.3).

Why the policies differ so little. In a generational EA with two elites per island, an island’s worst individuals are not random genomes: they are fresh children of tournament-selected parents, carrying the source island’s working neurons, and every policy replaces the receiver’s worst individuals. Recombined with the natives’ neurons (crossover probability 0.9), any emigrant injects some useful material; the best only spread it faster. The interval was not the reason: at every 20 generations the islands stayed distinct (one shared champion in 4% of the generations, against 28% at every 10), yet whom they sent mattered little. How often they migrate is the stronger lever: every 50 generations or never was clearly slower than 5–20.

Why one population did as well. Our budget is short (166 generations), and the panmictic population keeps the 8 best individuals overall, where the islands keep the 2 best of each island. At this budget a single population was as good as the islands on average, but not as dependable: in final fitness it was the best of the five EAs on 9 of 20 seeds and the worst on 7, while random migration was never the worst. One population either finds a good gait or gets stuck on its arena; islands hedge against getting stuck, the classic case for island models, and pay with fewer big wins. In the preliminary study the standard EA had ended best of all; with 20 seeds that lead disappeared, a reminder of how wide five-seed intervals are. [TODO: longer run, or future work?]

Limitations. The policy effects we found are small ($|d_z| \leq 0.61$) and none survives the correction for 15 comparisons; detecting them reliably would need more than 20 seeds. The settings were tuned with one factor at a time on three seeds, which misses interactions [8], and with the random policy, so they may suit the policies unevenly. Choosing the test by a normality check is a two-stage procedure; we fixed the rule in advance. The fitness shapes the gait (Section 2.3); the plain distance gives the same ordering (Table 4). Brains specialise to their arena, one body and world were used, and training on varied arenas did not help in a pilot [13].

6 CONCLUSION

We evolved neural controllers that make `spider_8` walk to a target 2 m away and asked whom an island model should send between its islands. With 20 paired seeds, every EA beats random search by far, but the emigrant-selection policy matters little: migrating the best or random individuals probably speeds up evolution compared with isolated islands, migrating the worst barely does, and no difference between the evolutionary algorithms is significant after correcting for multiple comparisons, in speed, final fitness or behaviour on unseen arenas; by end results a single population generalised best but was the least consistent, and random migration ended fittest. How often the islands migrate mattered more in a preliminary study: every 5–20 generations was equally fast and every 50 as slow as never. Future work: longer budgets, where island diversity should pay off; crossover that aligns an immigrant’s neurons with the natives’ [21]; and arenas varied enough for the brains to generalise. These conclusions rest on one body, one world and a fitness that shapes the gait.

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CONTRIBUTIONS

[TODO: each member: one or two lines on what they did]